

QRM II Graded Assignment (9)

Period 1, 2026

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0 Introduction

This assignment is to be completed in groups of 4-5. Further, all students in your group need to be assigned to the same R tutorial group (Friday's tutorial). You can sign yourself up for a group on Canvas. Please do so **before the start of your first R tutorial on Friday September 4th**. You can use the Discussion Board in Canvas if you do not have a group yet or if your group is incomplete.

The assignment has 5 parts, and each part corresponds to the course material of that week (with the exclusion of week 6, for which there is no R programming material).

You are supposed to hand in these assignments on Canvas at the following dates:

- **Deadline 1** *Sunday September 27th, at 23:59pm*: you are supposed to hand in weeks 1, 2, and 3 of this assignment. This will determine 18% of your overall course grade
- **Deadline 2** *Sunday October 11th, at 23:59pm*: you are supposed to hand in weeks 4, and 5 of this assignment. This will determine 12% of your overall course grade

The R tutorials (each Friday) will consist of two halves. During the first half, you will discuss the tutorial exercises. These can be downloaded separately from Canvas. During the second half, you can work on this graded assignment within your own group. The purpose is that you find out how to work with R for doing statistical analyses by yourself. The tutorial exercises are meant to teach you basic commands to get you started, but to answer the problem sets in this assignment, you might need to research your own solutions, and use functions and commands not described in the tutorial exercises. Learning how to solve your own research problems is integral part of learning R. When you and your group get stuck on how to approach an exercise, the hierarchy in finding your way is as follows:

- use the concepts from the tutorial exercises;
- use the cheat sheets available on Canvas;
- use Google, YouTube, StackOverflow, or another website;
- ask the teacher.

The use of generative AI is **not** permitted and may result in a grade of 0. See the AI protocol in the course manual for details.

To answer the assignment, you can simply fill out this R markdown document. There are designated places which you can fill with R code. There are also designated spaces for you to answer each question. Often, the structure of an answer will be as follows. First, you type the R code in the designated box. This will show how you analyzed the data to get the answer to the question. Below the box for the R code, you will then summarize your answer to the question, i.e. what are the conclusions that you draw from the data analysis?

When handing in, you are supposed to submit this .Rmd file, and a knitted version of this document. You can knit this document to pdf, word, or html. Knitting to pdf requires you to have a .tex distribution installed on your computer. Knitting to Word requires you to have Word installed.

The exercises are designed such that you should be able to finish the majority of them during the tutorial each week. If you are not able to finish them fully during that time, you are expected to work on it in your own time using the computers on campus or your own device. It is best to meet as a group in-person when working together. If you want to work remotely, github is a good platform to guarantee smooth collaboration. Alternatively, you can email this .Rmd file back and forth to one another as a group, but this is not recommended as it is more cumbersome.

We encourage you to keep your code blocks, printing statements, and final answers, as short as possible. In any case, there is a page limit of 6 pages per week, which encompasses the total length of this document which consists of the questions, your coding lines, and your answers. When your answers to questions of the respective week exceed this page limit, they will not be graded, resulting in zero points.

1 Week 1

1. Find the dataset “movies3.tsv” on Canvas. Describe your data set: How many observations does it have. How many variables are there? How many subjects? What consists of a subject? [4 points]

```
nrow(movies3)
```

```
## [1] 707
```

```
ncol(movies3)
```

```
## [1] 19
```

```
glimpse(movies3)
```

```
## Rows: 707
## Columns: 19
## $ index          <dbl> 2, 2460, 219, 2900, 2236, 386, 122, 4627, 1699, 46~
## $ budget         <dbl> 2.45e+08, 1.60e+07, 1.20e+08, 1.20e+07, 0.00e+00, ~
## $ keywords       <chr> "spy based on novel secret agent sequel mi6", NA, ~
## $ original_language <chr> "en", "en", "en", "ja", "en", "en", "en", "e~
## $ title          <chr> "Spectre", "The Unborn", "Charlie's Angels: Full T~
## $ popularity     <dbl> 107.376788, 15.788784, 33.616115, 5.158067, 13.621~
## $ release_date   <date> 2015-10-26, 2009-01-09, 2003-06-27, 2010-12-01, 2~
## $ revenue        <dbl> 880674609, 76514050, 259175788, 11212294, 0, 28850~
## $ runtime        <dbl> 148, 87, 106, 131, 117, 108, 107, 99, 104, 96, 104~
## $ vote_average   <dbl> 6.3, 4.8, 5.2, 6.3, 7.0, 7.2, 6.2, 6.2, 6.1, 5.7, ~
## $ vote_count     <dbl> 4466, 365, 914, 59, 216, 2825, 4021, 11, 400, 16, ~
## $ genre          <chr> "Action", "Thriller", "Comedy", "Fiction", "Drama"~
## $ release_year   <dbl> 2015, 2009, 2003, 2010, 2012, 2004, 2009, 2015, 20~
## $ release_month  <dbl> 10, 1, 6, 12, 10, 7, 4, 2, 4, 1, 2, 2, 3, 2, 4, 11~
## $ release_day    <dbl> 26, 9, 27, 1, 25, 23, 28, 7, 6, 21, 11, 13, 24, 3,~
## $ first_actor    <chr> "Daniel Craig", "Odette Annable", "Cameron Diaz", ~
## $ first_actor_gender <chr> "male", "female", "male", "male", "male", "male", ~
## $ director_first_name <chr> "Sam", "David", "McG", "Takashi", "Curtis", "Paul"~
## $ director_gender <chr> "male", "male", NA, "male", "male", "male", "male"~
```

Your Answer:

Observations: 707 Variables: 19 Subjects: 707 A subject is a movie entry in the dataset, represented by different attributes like, title, budget, revenue, release date, actors and director.

2. Which of the following types of variables are present in your data set? (i) nominal; (ii) ordinal; (iii); continuous; (iv) discrete. If present, name one example of such a variable present in your data set. [4 points]

```
str(movies3)
```

```
## spc_tbl_ [707 x 19] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ index          : num [1:707] 2 2460 219 2900 2236 ...
## $ budget         : num [1:707] 2.45e+08 1.60e+07 1.20e+08 1.20e+07 0.00 7.50e+07 1.50e+08 0.00 ...
## $ keywords       : chr [1:707] "spy based on novel secret agent sequel mi6" NA "robbery secret ...
## $ original_language : chr [1:707] "en" "en" "en" "ja" ...
## $ title          : chr [1:707] "Spectre" "The Unborn" "Charlie's Angels: Full Throttle" "Space ...
## $ popularity     : num [1:707] 107.38 15.79 33.62 5.16 13.62 ...
## $ release_date    : Date[1:707], format: "2015-10-26" "2009-01-09" ...
## $ revenue        : num [1:707] 8.81e+08 7.65e+07 2.59e+08 1.12e+07 0.00 ...
## $ runtime        : num [1:707] 148 87 106 131 117 108 107 99 104 96 ...
## $ vote_average    : num [1:707] 6.3 4.8 5.2 6.3 7 7.2 6.2 6.1 5.7 ...
## $ vote_count      : num [1:707] 4466 365 914 59 216 ...
## $ genre          : chr [1:707] "Action" "Thriller" "Comedy" "Fiction" ...
## $ release_year    : num [1:707] 2015 2009 2003 2010 2012 ...
## $ release_month   : num [1:707] 10 1 6 12 10 7 4 2 4 1 ...
## $ release_day     : num [1:707] 26 9 27 1 25 23 28 7 6 21 ...
## $ first_actor     : chr [1:707] "Daniel Craig" "Odette Annable" "Cameron Diaz" "Hiroyuki Ikeuchi ...
## $ first_actor_gender : chr [1:707] "male" "female" "male" "male" ...
## $ director_first_name : chr [1:707] "Sam" "David" "McG" "Takashi" ...
## $ director_gender  : chr [1:707] "male" "male" NA "male" ...
## - attr(*, "spec")=
## .. cols(
## ..   index = col_double(),
## ..   budget = col_double(),
## ..   keywords = col_character(),
## ..   original_language = col_character(),
## ..   title = col_character(),
## ..   popularity = col_double(),
## ..   release_date = col_date(format = ""),
## ..   revenue = col_double(),
## ..   runtime = col_double(),
## ..   vote_average = col_double(),
## ..   vote_count = col_double(),
## ..   genre = col_character(),
## ..   release_year = col_double(),
## ..   release_month = col_double(),
## ..   release_day = col_double(),
## ..   first_actor = col_character(),
## ..   first_actor_gender = col_character(),
## ..   director_first_name = col_character(),
## ..   director_gender = col_character()
## .. )
## - attr(*, "problems")=<pointer: 0x000001df054fca50>
```

Your Answer:

There are nominal variables: “title” is an example. There are no ordinal variables. There are continuous variables: “budget” is an example. There are discrete variables: “release_year” is an example.

3. A movie studio wants to know which types of movies give maximal revenue. Perform the following steps to provide the movie studio with an analysis which corresponds to their request:
 - a. Summarize the variable revenue and report its mean, median, maximum, and minimum. **[2 points]**
 - b. Which movie has the highest revenue in your data set and how much is this revenue. Which movie has the lowest and how much is its revenue? If multiple movies have the exact same highest or lowest revenue, give only one example. **[2 points]**
 - c. Create a histogram of the variable revenue. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Describe the shape of the histogram. What does this tell you about the nature of making money in the movies industry? **[2 points]**
 - d. Add a new variable to your data set: the log of revenue. When creating this variable, what happens to movies for which revenue is zero? What then happens when you calculate the mean of log revenue? **[2 points]**
 - e. For movies that have a revenue of zero, replace log of revenue with “NA”. What is now the mean of log of revenue? Create a histogram for log revenue, again with an appropriate title, x- and y-axis labels. Describe the shape of this histogram. **[2 points]**
 - f. For the variables revenue and log of revenue calculate their skewness and kurtosis. How do these numbers relate to the shape of the histograms that you estimated under d.) and e.)? **[2 points]**

For each step, you should provide first all the code you used to answer the question and then formulate an answer using full sentences.

Each week consists of 1, 2, or 3 subquestions. The total amount of points you can earn per week is 20 points.

Step a

```
mean(movies3$revenue, na.rm = TRUE)
```

```
## [1] 77277126
```

```
median(movies3$revenue, na.rm = TRUE)
```

```
## [1] 17986781
```

```
max(movies3$revenue, na.rm = TRUE)
```

```
## [1] 1153304495
```

```
min(movies3$revenue, na.rm = TRUE)
```

```
## [1] 0
```

Your Answer:

The variable revenue has a mean of \$77,271,126, a median of \$17,986,781, a maximum of \$1,153,304,495, and a minimum of \$0.

Step b

```
movies3[which.max(movies3$revenue), c("title", "revenue")]
```

```
## # A tibble: 1 x 2
##   title                      revenue
##   <chr>                     <dbl>
## 1 Captain America: Civil War 1153304495
```

```
movies3[which.min(movies3$revenue), c("title", "revenue")]
```

```
## # A tibble: 1 x 2
##   title                      revenue
##   <chr>                     <dbl>
## 1 Chasing Mavericks          0
```

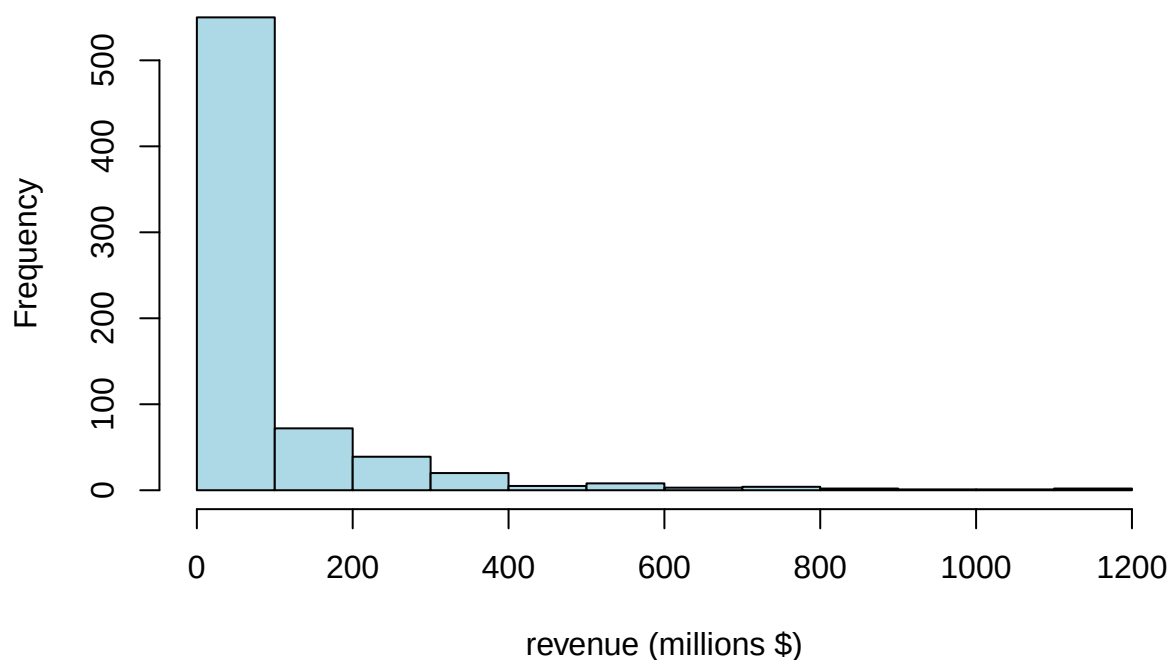
Your Answer:

The movie with the highest revenue is Captain America: Civil war and the revenue is \$1,153,304,495 and one of the movies with the lowest revenue is Chasing Mavericks with a revenue of \$0.

Step c

```
hist(movies3$revenue / 1e6,
      main = "Histogram movie revenue",
      xlab = "revenue (millions $)",
      ylab = "Frequency",
      col = "lightblue")
```

Histogram movie revenue



Your Answer:

You can see that the histogram is right-skewed. A high percentage of movies make relatively lower revenue than a very small percentage that generate a very high revenue.

Step d

```
movies3$log_revenue <- log(movies3$revenue)
mean(movies3$log_revenue, na.rm = TRUE)
```

```
## [1] -Inf
```

Your Answer

The answer is -Inf, that is because there are movies with zero revenue. And taking the logarithm of 0 yields negative infinity, because $\ln(0)$ is undefined. When calculating the mean, these $\ln(0)$ values bring the entire sum down.

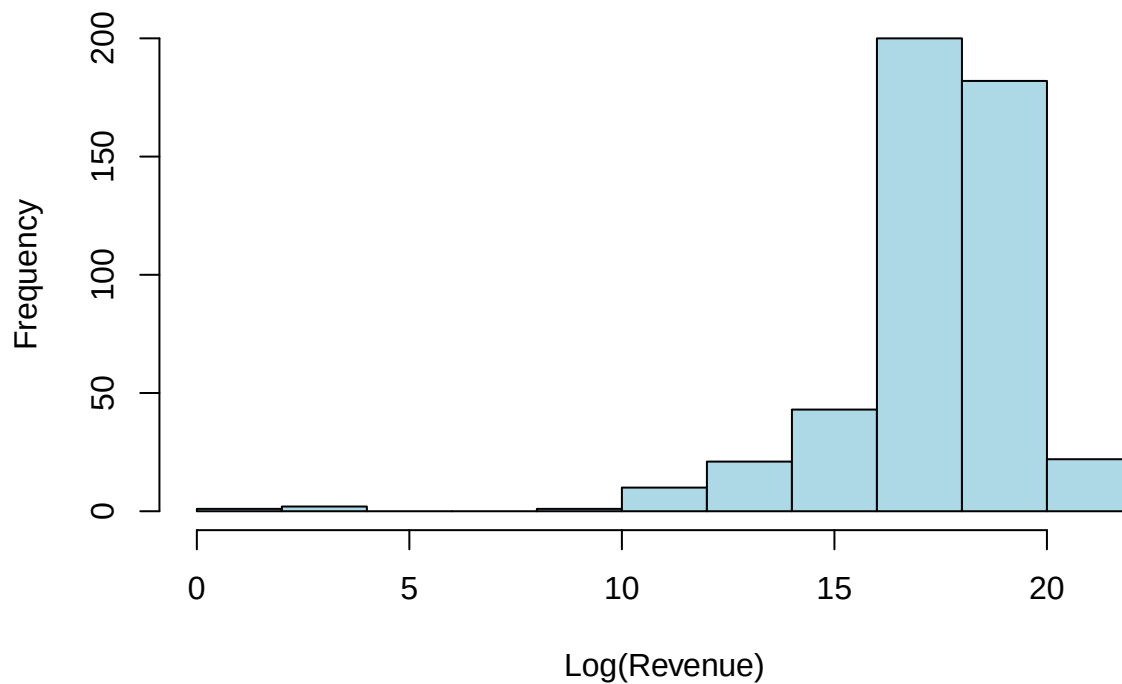
Step e

```
movies3$log_revenue[is.infinite(movies3$log_revenue)] <- NA
mean(movies3$log_revenue, na.rm = TRUE)
```

```
## [1] 17.35027
```

```
hist(movies3$log_revenue,
     main = "Histogram movie revenue",
     xlab = "Log(Revenue)",
     ylab = "Frequency",
     col = "lightblue")
```

Histogram movie revenue



Your Answer:

The mean is now 17.35027. The histogram is left skewed and the most movies hover around a value of 16. And there are a few with lower revenues.

Step f

```
library(moments)
```

```
skewness(movies3$revenue, na.rm = TRUE)
```

```
## [1] 3.553064
```

```
kurtosis(movies3$revenue, na.rm = TRUE)
```

```
## [1] 18.75095
```

```
skewness(movies3$log_revenue, na.rm = TRUE)
```

```
## [1] -2.469978
```

```
kurtosis(movies3$log_revenue, na.rm = TRUE)
```

```
## [1] 14.52609
```

Your Answer:

The variable revenue has a skewness of 3.55 and a kurtosis of 18.75. In the histogram we see the same movement so these numbers confirm what we can witness in the histogram, a tail to the right.

The variable log_revenue has a skewness of -2.47 and a kurtosis of 14.53. The negative skewness is also noticeable in the histogram. The tail is now on the left and the distribution is less extreme than with the original values.

2 Week 2

1 Is your dataset movies3.tsv the full population, or is it a sample of a larger population? If the latter, how would you describe the full population? [4 points]

Your answer here:

It is a sample of a larger population, because it is not every movie that has ever been made. If it was the full population then it would have been every movie ever.

2 For this question, you will assume that your data set is the full population.

- What is the mean of the variable vote_average in your data set? [2 points]
- Create a new data set, called movies_sample. Make sure that it is a random sample of your data set of 25 movies. What is the mean of vote_average in this random sample? How does it compare to the mean of vote_average in 2a? [2 points]
- In a for loop, create 100 different samples of 25 movies, as in b, and estimate the mean within each sample. Save the mean of each sample in a vector called sample_means. So the first position of the vector would have the mean of the first sample, the second position the mean of the second sample, etc. Print the start of this vector. [2 points]
- Summarize and make a histogram of sample_means. What is the mean, standard deviation and shape of its distribution? [2 points]
- Using the formulas of the *Central Limit Theorem*, give the distribution of the sample mean of vote_average in a sample of 25 movies from your population (including its expected value and standard deviation). How do these values compare to your answers at d.)? [2 points]

Step a

```
mean(movies3$vote_average, na.rm = TRUE)
```

```
## [1] 5.971994
```

Your Answer:

The mean of the variable vote_average is 5.97.

Step b

```
set.seed(123)
movies_sample <- movies3[sample(1:nrow(movies3), 25), ]
mean(movies_sample$vote_average, na.rm = TRUE)
```

```
## [1] 6.228
```

Your Answer:

The mean of the variable `vote_average` of this random sample is higher than the average of the whole dataset, that is due to random sampling variability.

Step c

```
sample_means <- numeric(100)
for (x in 1:100) {
  sample_1 <- movies3[sample(nrow(movies3), 25), ]
  sample_means[x] <- mean(sample_1$vote_average, na.rm = TRUE)
}
print(head(sample_means))
```

```
## [1] 6.024 5.596 6.100 6.020 6.292 6.108
```

Your Answer:

Because it is a random sample, we cannot write down the first numbers as an answer because they are different every time the code is ran.

Step d

```
#WRITE YOUR CODE HERE
```

Your Answer:

Write your formulated response here.

Step e

Your Answer:

Write your formulated response here.

3. Is there a variable in your data set which you would expect to follow a discrete uniform distribution? Explain, without looking at the data, why you expect this data to be discrete uniform. If the data would indeed follow a discrete uniform distribution, what would you expect the mean and standard deviation of this variable to be? Next, plot a histogram of the variable, and plot an appropriate discrete uniform distribution on top of this histogram. Comment whether the variable indeed follows this expected distribution. If the distribution deviates from the discrete uniform distribution, explain why this is the case. [6 points]

Your Answer:

Write your formulated response here.

```
#WRITE YOUR CODE HERE
```

Your Answer:

Write your formulated response here.

3 Week 3

For the next part of the assignment, assume that the movies in your data frame are a random sample of a larger population of movies.

1. Give a 90 percent confidence interval for the variable *revenue*. Interpret the result. [2 points]

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

2

- a. How many movies in your sample were directed by a male director? How many by a female director? [2 points]
- b. Test the null hypothesis that the chance that a movie is being directed by a male or female director is equal. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Use an appropriate R function to test your hypothesis, and state your conclusion. Also give the p-value that corresponds to your test statistic. [2 points]
- c. Recode revenue such that it is measured in millions of dollars. What is the variance of revenue for movies with a male director? What is the variance of revenue for movies with a female director? [2 points]
- d. Assume that the variance of revenue for movies with a male director in your data is the same as the variance for movies with a male director in your population (that is, there is no sampling uncertainty in this estimate). Now, set up a hypothesis test to test for the null hypothesis that the variance of revenue for movies with a female director is equal to the variance for movies with a male director. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Calculate your test statistic and tell how your test statistic is distributed, and state your conclusion. Also give the p-value that corresponds to your test statistic. [2 points]

step a

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step b

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step c

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step d

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

3

- a. Create a scatter plot with year on the x axis and mean of revenue within that year on the y-axis. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Is revenue of movies increasing or decreasing over time? **[3 points]**
- b. Recreate the scatter plot with year on the x axis and mean_revenue on the y-axis, but now add bars around each point, indicating the 95% confidence interval. **[3 points]**
- c. If you look closely at the resulting plot in b.), you will probably see that the confidence intervals in some years are wider than in other years. What are some possible reasons that a confidence interval in a given year can be wide? Can you verify this using your data? **[4 points]**

step a

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step b

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step c

#WRITE YOUR CODE HERE

4 Week 4

1. There is an argument going on in the movie studio. *Bob* claims that production budgets are getting out of hand, and that the studio should focus on making cheaper movies. *Chantal* disagrees. She tells Bob that “Every dollar we spend on movie production is more than offset by the increase in movie revenue’”.
- a. Set up a regression model to test Chantal’s claim, and estimate it. That is, estimate:

$$\text{Revenue}_i = \beta_0 + \beta_1 \text{Budget}_i + \varepsilon_i.$$

Print a summary of your estimated model. **[2 points]**

- b. What is the estimated value of β_1 and how do you interpret it? [2 points]
- c. Test for the null hypothesis that $\beta_1 \leq 1$. Report the p-value and state your conclusion. [2 points]
- d. Next, estimate the model

$$\text{Log Revenue}_i = \beta_0 + \beta_1 \text{Log Budget}_i + \varepsilon_i.$$

When creating the variables Log Revenue and Log Budget, make sure that movies with a Revenue or Budget of zero are assigned the value “NA”. Print a summary of your estimated model. [2 points]

- e. What is the estimated value of β_1 and how do you interpret it? [2 points]
- f. Which model has better fit? The level-level model or the log-log model? Explain. [2 points]
- g. Who do you think is correct? Bob or Chantal? What would you advise the movie studio to do? [2 points]

step a

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step b

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step c

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step d

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step e

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step f

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step g

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

2

- a. Make a scatterplot with log revenue on the y-axis, and log budget on the x-axis. Within the same scatterplot, draw a straight line which summarizes the association between the log of revenue and log of budget. **[3 points]**
- b. Using the insights from the scatterplot, your linear model, and your data, find an example of a movie that vastly overperformed. That means, find a movie that had much more revenue than expected, given its budget. For this movie, report its log of revenue, its predicted log of revenue, and its residual. **[3 points]**

step a

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step b

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

5 Week 5

- a. Make a frequency table of the variable genre. Which are the three genres that occur the most? **[2 points]**
- b. Create a new variable called genre2. Make sure that the three most popular genres remain the same, but that all the other genres get categorized into one genre called “Other”. Report a frequency table of this new variable genre2. **[2 points]**
- c. Estimate an OLS model which has as dependent variable the log of revenue of a movie, and as independent variable the log of budget, a dummy for each level that genre2 can take, and the year of release. Show a summary of the resulting model and interpret each coefficient. **[4 points]**

The movie studio that you work at is releasing a new movie in 2026. It will be a Drama movie with a budget of 20,000,000. It will have a runtime of 120 minutes.

- d. Estimate a model that is able to predict the revenue of this movie. Give its predicted revenue and include a 90% prediction interval. [5 points]
- e. Plot the residuals against various independent variables in the model. Do you find any evidence of heteroskedasticity? [2 points]
- f. One executive at the studio wants to time the release of the movie to a specific month of the year such that they can maximize revenue. Another executive is of the opinion that the month in which the movie is released will not have a big impact on its revenue. Estimate a model that can test whether the month of release impacts the revenue of a movie. Use an appropriate hypothesis test to test whether month of release matters. What would you advise the movie studio regarding the timing of the release of the movie? [5 points]

step a

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step b

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step c

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step d

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step e

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step f

#WRITE YOUR CODE HERE